

# Understanding AHRI's Telephone Activity

A Power BI dashboard for inbound demand and outbound activity

Aji Pratama

Sumintra Boonmat

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ETC5543 Business analytics creative activity — Group project report

Git repository: [AHRI Call Dashboard Project](#)

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## 1 Background and objectives

### 1.1 The organisation and the reporting problem

The Australian HR Institute (AHRI) is a professional membership body supporting human resources practitioners. Its activities include professional certification, training, practical resources and events. These services generate different reasons for members and prospective members to contact the organisation, from membership enquiries to questions about professional

development ([Australian HR Institute, n.d.](#)). Phone activity is therefore part of service delivery, alongside email and online enquiries. Understanding this activity matters because the same staff may need to manage several channels during a working day.

At the first meeting, AHRI described a gap between its digital and telephone reporting. Email and online enquiries fed into its Dynamics customer relationship management system, where existing dashboards provided visibility. Telephone activity was handled through Microsoft Teams, but the Member Engagement Team relied on raw exports obtained by IT. Staff could inspect individual records, yet there was no convenient shared view of demand, destinations or calling patterns. The project addressed that reporting gap rather than proposing a replacement telephone system ([AHRI and Monash University project team, 2026](#)).

The initial brief asked for call volumes, routing destinations, answer and abandonment rates, waiting time and talk time. It also identified outbound activity as a possible extension. However, a requested metric is not automatically a measurable one. The files eventually available contained technical call records rather than a complete record of queue events or customer conversations. The project therefore had to establish what each row represented before selecting indicators. This distinction became the central analytical challenge, because incorrect counting could produce a visually attractive but misleading dashboard.

The intended audience is AHRI's Member Engagement Team and colleagues responsible for understanding telephone operations. They need a report that supports questions such as when inbound demand is concentrated, which services appear in queue records, and how outbound activity varies across teams. They also need clear boundaries around interpretation. A technical connection status is useful information, but it cannot establish whether a member spoke to an employee or had their enquiry resolved. The dashboard is designed to support investigation and planning without presenting unsupported service-performance claims.

## 1.2 How calls move through the system

An inbound caller may contact AHRI's main telephone number and hear an automated menu, commonly called an interactive voice response system, or IVR. The caller's choice can direct the call towards a service such as Membership or Events. A call queue is a system component that distributes calls to staff according to its configuration. An overflow destination is an alternative destination configured for particular queue conditions. Microsoft distinguishes queue overflow, timeout and other exception settings; the export alone does not reveal which configured condition actually operated for a particular record ([Microsoft, n.d.](#)).

One telephone interaction can produce several records as it moves between system accounts. For example, a person may enter the main menu and then reach a queue. The export can contain a record associated with the menu and another associated with the queue. Counting both as separate incoming customer enquiries would inflate demand. Conversely, counting only rows labelled Main IVR when examining routing would discard related records that help describe the destinations observed. The project uses separate record-level and group-level views to address these different questions.

There is also a route that bypasses the main menu: someone may call a staff member directly. During the project, Vanessa confirmed that calls without a Main IVR record were mostly direct calls to staff. This clarification explains why the export contains more linked groups than Main IVR entries. It does not prove that every remaining group has the same explanation. The report consequently uses the observed label "without a Main IVR record" when discussing the full reconciliation, while using the stakeholder explanation to justify the narrower inbound focus.

### 1.3 Objectives and contribution

The first objective was to establish defensible counting rules for inbound demand and outbound activity. This required distinguishing source rows, related-record groups, Main IVR entries and outbound attempts. The second objective was to prepare a consistent model that could support date, service, team and staff comparisons. The third was to build clear Power BI pages through which users could explore those comparisons. The fourth was to evaluate the reliability and limits of the resulting measures, including the effects of incomplete dates, missing fields and ambiguous technical outcomes.

The contribution is a practical analytical tool supported by documented definitions and checks. It combines Power Query preparation, a structured semantic model, DAX calculations and interactive report pages. The value comes from making existing records understandable and reusable rather than developing a new statistical algorithm. A particularly useful contribution is the reconciliation of apparently conflicting call totals. Explaining why several thousand source rows reduce to a smaller number of relevant groups gives users a sound basis for interpreting every subsequent chart.

The scope is retrospective visibility over the supplied export period. Real-time monitoring, service-level targets, customer satisfaction, staff productivity and causal evaluation are outside the current scope. Forecasting was also excluded: the immediate priority was reliable measurement, and the available period is too limited to justify strong claims about recurring annual patterns. This report evaluates the repository snapshot dated 6 October 2026. It reports verified results alongside implementation limitations, and separates proposed improvements from work already completed.

## 2 Data sources and initial data analysis

### 2.1 Files, coverage and analytical populations

The project uses two Excel files supplied for AHRI's Microsoft Teams Direct Routing activity. The inbound workbook, `DirectRouting_Ehsan_2026-09-01.xlsx`, contains 9,859 inbound records. The later workbook, `Export_Outbound.xlsx`, contains 17,655 records covering both directions: 9,859 inbound and 7,796 outbound. Despite its filename, the second workbook is not restricted to outbound calls. Filtering Call Direction is therefore necessary before analysing outbound activity. Comparing the common columns also confirms that its inbound records match those in the original workbook.

The observed overall coverage runs from 6 April to 1 September 2026 in Melbourne time. These dates indicate the earliest and latest observed activity, but do not confirm that both boundary days were fully captured. For example, an export could end partway through its final day. The model therefore excludes the first and last observed dates when calculating daily averages. It treats May through August as fully covered months, assuming that the export contains continuous coverage between its boundaries. This assumption needs confirmation when another export is added.

The main reporting populations are summarised in Table 1. These counts describe different levels and subsets of the data, so they should not be added together.

Table 1: Source populations and counting units, recalculated from the workbooks stored in the project repository.

| Reporting population              | Count | Meaning                                        |
|-----------------------------------|-------|------------------------------------------------|
| Original inbound records          | 9,859 | Individual system records                      |
| Distinct inbound shared-ID groups | 6,361 | Groups of related records                      |
| Groups containing Main IVR        | 3,986 | Main-menu population used for inbound analysis |
| Other inbound groups              | 2,375 | Groups without a Main IVR record               |
| All outbound records              | 7,796 | Outbound records before staff selection        |
| Selected outbound attempts        | 6,046 | Attempts associated with the approved staff    |

The outbound analysis follows the stakeholder’s list of staff in Membership, HRSC and Events. It includes eleven Membership staff, five HRSC staff and two Events staff. Their account-to-team mapping is stored in `DimStaff`, using the user principal name, or UPN, as the matching key. Name variations involving Lara and Georgina were clarified during the project before finalising the mapping. Matching against the approved accounts excludes 1,750 outbound records outside the agreed staff list. These records are outside the scope of this analysis; their exclusion does not mean they are incorrect.

## 2.2 Understanding the main fields

The two main identifiers are `CorrelationId` and `Shared correlation ID`. `CorrelationId` is a technical identifier associated with a recorded call. `Shared correlation ID` connects related call records. Neither identifies a unique person. The UPN identifies the associated user or resource account, while `Display Name` provides a readable account label. `Call Direction` distinguishes inbound from outbound activity. `Invite time` records the initial call invitation and provides a timestamp for analysing attempts, including unsuccessful ones. `Duration`, `Success`, `SIP codes` and `SIP phrases` describe recorded technical activity ([Microsoft, 2025, 2026a](#)).

These fields need to be interpreted at the correct level. A shared-ID group can contain several rows associated with different accounts or destinations. It should not automatically be treated as a complete customer journey, because the export may not contain every event needed to reconstruct that journey. A telephone number also cannot establish the identity of a unique customer. Numbers may be shared, withheld or unavailable. The report therefore avoids presenting shared-ID groups as unique people or using these records to make claims about individual customer behaviour.

## 2.3 Missing values and their treatment

Initial data analysis checked the number of records, call directions, missing values, duplicate rows, repeated identifiers and timestamp coverage. The inbound workbook contains 334 missing UPN values and 334 missing display names. It also contains 785 records without start and end timestamps. Records with missing account information were retained, with their account identity treated as unknown. Other fields in these rows can still contribute to the analysis,

including shared identifiers, invitation timestamps and technical outcomes. Removing the rows would change the reporting population without establishing that the records were invalid.

Missing timestamps were also retained rather than replaced with estimated values. In the inbound workbook, all 785 records without start timestamps have an unsuccessful status. In the selected outbound population, all 542 unsuccessful attempts have missing start timestamps and zero recorded duration. These patterns show why missing values should be considered alongside other fields. Replacing a missing start time with Invite time could imply that a connection occurred when the source does not establish one. Deleting unsuccessful attempts would also understate activity and remove information needed to describe technical outcomes.

The preparation therefore keeps these records while applying the relevant population filter to each measure. Outbound attempt counts include both technically successful and unsuccessful records. Successful-duration measures use records marked successful. Missing account details remain a limitation when identifying destinations, but do not automatically make every other field in the same row unusable. This approach preserves the available evidence without filling gaps with unsupported assumptions.

## **2.4 Duplicate checks and consistent data types**

No exact duplicate rows were identified in the inbound workbook. However, one correlation identifier appears more than once. This distinction matters because a repeated identifier does not necessarily mean that the entire row is duplicated. Records must be compared before deciding whether anything should be removed. Two inbound columns contain no observed values: an unnamed Excel column and Administrative Units. These columns are excluded during preparation because they contribute no information to the current analysis.

Phone numbers and identifiers are stored as text rather than numeric values. Telephone fields can contain values such as `anonymous`, and numeric conversion can also remove leading zeros or mishandle international prefixes. Duration is assigned a numeric type, while dates and timestamps receive the appropriate temporal types. Account text is cleaned, trimmed and converted to lowercase before matching. These preparation steps reduce avoidable matching errors and provide consistent fields for the relationships and calculations described in the next section.

# **3 Analytical approach and Power BI implementation**

## **3.1 Separating data preparation from calculations**

The implementation separates data preparation, model relationships and reporting measures. Power Query imports the workbooks, cleans fields, selects the relevant records and creates classification columns. Relationships connect the prepared tables so that date, team and staff selections filter the appropriate records. Data Analysis Expressions (DAX) measures then calculate the values shown in the report. This separation makes it easier to investigate an unexpected result: the issue may come from the source data, a preparation step, a relationship or the measure itself.

## 3.2 Identifying Main IVR entries and related records

The inbound model starts with `FactDirectRouting`, which contains the prepared inbound records. Power Query renames the source UPN field to `ResourceAccount`. A record is classified as a Main IVR entry when the cleaned account equals `aa-mainivr-v2@ahri.com.au` and its `CorrelationId` equals its Shared correlation ID. The account condition identifies the main-menu resource, while the identifier condition selects the entry record used by the measure. In this export, the rule produces 3,986 entries.

A second table, `FactRoutingGroup`, groups the inbound records by Shared correlation ID. Each group receives indicators showing whether it contains Main IVR, queue, overflow-named, user-associated or unsuccessful records. Its entry date is the earliest observed Invite time within the group, converted to Melbourne time. The resulting table has one row per related-record group. This supports group-level summaries while retaining the individual records in `FactDirectRouting` for more detailed analysis.

The two tables answer different questions. Counting source rows describes recorded system activity. Counting shared-ID groups combines related records. Restricting the groups to those containing Main IVR then applies the agreed inbound scope. The reduction from 9,859 records to 6,361 groups is therefore a grouping operation, while the reduction to 3,986 Main IVR groups is a scope decision. Neither step should be described as removing thousands of duplicate or invalid calls.

The Main IVR entry count equals the number of groups containing the Main IVR account in this export. This agreement provides a useful check, but it should be tested again with future data. The model does not assume that every possible export will contain exactly one qualifying entry per group. Keeping separate entry and group measures makes their counting rules visible and allows differences to be investigated.

## 3.3 Classifying queue destinations

Queue classification uses the presence of “Queue” in Display Name. A queue record is additionally classified as overflow when its name contains “Overflow”. Service names are derived from these labels so that primary and overflow destinations can be compared within the same service. This approach uses information available in the export, but depends on consistent account naming. A maintained account-to-service mapping would be more reliable if names or routing arrangements change.

Main IVR groups are assigned to three mutually exclusive categories. A group containing an overflow-named queue record is classified as “Overflow queue observed”. A group containing queue records but no overflow-named record is classified as “Primary queue only observed”. The remaining groups are classified as “No queue record observed”. These categories describe the available records. The absence of a queue record does not establish that the caller abandoned the menu or that the call ended at that point.

The queue measure counts distinct shared IDs associated with queue records in the selected service and period. Overflow destination share divides groups containing overflow-named records by groups containing any queue record. Primary and overflow counts can overlap when the same group contains both types of record. The percentage therefore describes recorded destination associations, rather than a confirmed probability of escalation.

The percentage chart displays a service only when it has at least thirty queue-associated groups in the current selection. Here, `n` means the number of observed groups, and the rule is `n >= 30`. Thirty is a display threshold chosen during development, not a value extracted from Excel or a

telephone queue capacity. It reduces attention to percentages based on very small counts, but does not guarantee statistical reliability. Its sensitivity is considered in the evaluation section.

### 3.4 Selecting and counting outbound activity

Outbound preparation starts from `RawCallLogs`, filters Call Direction to Outbound and standardises the UPN before matching it against `DimStaff`. An inner join retains records associated with the approved staff accounts. The nested match column is then removed because staff names and teams remain available through the model relationship. The mapping contains one row per UPN, supporting an unambiguous account-to-team assignment and the one-to-many relationship required by the model.

The resulting `FactOutbound` table contains 6,046 records and 6,046 distinct correlation IDs. The main outbound measure uses a distinct count of `CorrelationId` to make its counting unit explicit. This equals the row count in the current selected dataset. A separate measure counts distinct staff accounts with recorded outbound activity. That measure describes observed participation within the selection, rather than the number of employees available or rostered to work.

For example, selecting one week may reduce the staff count because some approved staff have no recorded attempts during that week. Those staff remain in the approved mapping, but do not appear in the activity-based count. This distinction prevents the dashboard from presenting a changing activity count as a changing team headcount. Team and staff selections are applied through `DimStaff`, allowing the same measures to be reused consistently across different views.

### 3.5 Time conversion and average activity

Invite timestamps are parsed as UTC and converted to Melbourne time before the date and hour fields are created. The Power Query logic uses date-based daylight saving rules to select an offset of ten or eleven hours. The supplied reporting period falls within the ten-hour offset. Extracting the date after conversion is important because an evening UTC timestamp can fall on the following calendar day in Melbourne. Using the original UTC date would therefore misclassify some daily and hourly activity.

The date table contains every date between the observed inbound boundaries. Average weekday activity is calculated over selected Monday-to-Friday dates that pass the complete-day rule. Dates with zero activity remain in the denominator, while the first and last observed dates are excluded. Public holidays remain included because the model does not contain a verified AHRI operating calendar. The measure should therefore be understood as an average per calendar weekday, rather than per confirmed working day.

Across the full export, there are 105 eligible weekdays. These contain 3,925 Main IVR entries and 6,027 selected outbound attempts. Dividing each count by 105 gives 37.4 inbound entries and 57.4 outbound attempts per weekday. These numerators differ from the full-period totals because the averages use a narrower set of dates. The monthly weekday comparison applies the same principle, allowing months with different numbers of weekdays to be compared more fairly.

The model also defines a trailing seven-day average. For each date, it divides the Main IVR entries from that date and the preceding six calendar days by seven. Each window includes one occurrence of every weekday and retains zero-activity days. This reduces short-term variation while preserving the broader movement in activity. Only windows entirely within the accepted source boundaries are included. The calculation can retrieve preceding dates outside a narrow date selection, so selecting one date does not automatically reduce its window to one day.

### 3.6 Technical outcomes and duration

Technical outcome measures use the source’s Yes and No values. The outbound technical success rate divides successful attempts by all selected attempts. Mean and median duration are calculated from technically successful outbound records, including recorded zero durations. The mean describes the arithmetic average, while the median identifies the middle observation and is less affected by unusually long durations. These measures describe recorded technical activity; they are not treated as staff talk time or evidence that a customer’s enquiry was resolved.

### 3.7 Relationships and reproducibility

The date table, `DimRoutingDate`, has active relationships with the date fields in `FactDirectRouting`, `FactRoutingGroup` and `FactOutbound`. `DimStaff` connects to `FactOutbound` through `UPN`. Filtering operates from the dimension tables towards the fact tables. This follows the dimensional modelling principle of separating attributes used for filtering from observations used for summarisation (Microsoft, 2024). During development, a missing outbound date relationship prevented daily filters from reaching the outbound records. Creating that relationship corrected the weekday-average behaviour.

Both workbook queries use the `ProjectRoot` parameter to construct their source paths. Collaborators can therefore update their local project location without rewriting each query. The Power BI project format stores report and model definitions in text files, supporting version control and inspection of changes (Microsoft, 2026b). Separate checks against the Excel workbooks provide an additional comparison for the main counts. These checks support the intended calculation rules, but do not replace testing the rendered report and its interactive filters in Power BI Desktop.

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